

INNOVATION BY FACULTY IN TEACHING, RESEARCH AND DEVELOPMENT

- Integration of technology in teaching such as virtual labs, multimedia presentations, animations, and online platforms (Google Classroom, Microsoft Teams, LMS) to create interactive and accessible learning environments.
- Use of innovative pedagogical methods like Project-Based Learning (PBL), Case-Based Learning, flipped classrooms, role plays, group discussions, and seminars to foster critical thinking, creativity, and self-directed learning.
- Development and sharing of digital resources such as lecture notes, lab manuals, video tutorials, and YouTube channels for extended learning beyond classroom hours.
- Adoption of modern assessment techniques including practical-based assignments, quizzes, peer reviews, and continuous evaluation strategies to better measure student understanding and skills.
- Encouragement of faculty research involving students in research projects, publications in high-impact journals, and participation in national/international conferences.
- Collaboration with industry experts for co-teaching, guest lectures, and real-world exposure to current technologies and trends.
- Innovations in laboratory teaching through demonstration models, animations, and pre-lab instructional videos to enhance practical understanding.
- Emphasis on social responsibility, digital libraries, and blended learning approaches to cater to diverse student needs.
- Incorporating innovative teaching methods such as quizzes, debates, fashion shows, and cultural events to engage students actively.
- Encouraging research and development through participation in national-level events like NAUGHTICA (acoustic band competitions) and Navgati (maritime cultural festival).
- Achieving recognition through awards in innovation and academic excellence, e.g., IIM Innovation Award, Nautical Quiz, Seamanship Skills, and dance and paper presentations.
- Faculty and students collaborate on projects showcased in events at institutions like IMU Chennai, VIT Chennai, and others.
- Involvement in skill-building activities at national camps such as NCC RDC, fostering leadership and discipline alongside academics.
- Sports and fitness achievements in events like the Coast Guard Fitness Festival, emphasizing holistic development.

INNOVATION BY FACULTY IN TEACHING

EXPERIENTIAL LEARNING

To promote active learning, skill development, and higher engagement among Nautical Science students at the Academy of Maritime Education and Training (AMET), the following student-centric methods are followed:

1. Experiential Learning

Hands on Training at Simulator Lab



IALA Buoyage system was checked along with control of an RC vessel



Personal Survival Techniques



Flipped Classroom

- Students view instructional content (videos, readings) before class.
- Class time is used for discussions, activities, and problem-solving.

Group Discussion (Academics)



Industrial Visits & Port/Shipyards Exposure

- Field trips to ports, shipyards, and VTS centres.



Enhances practical understanding of theoretical concepts.



Participative Learning

Task Proficiency Demonstration (Academic)



Task Proficiency Demonstration (Practical)



Students experiencing to use sextant instrument to determine the angle between the horizon and a celestial body like the Sun, Moon or Star. Used in celestial navigation to find out the latitude and longitude.



A crew of cadets roaring a boat (Team Work)



Case Study Approach

- Utilizes real incidents (e.g., MV Wakashio grounding) for analysis.
- Promotes understanding of safety, regulations (SOLAS, MARPOL, ISM Code), and ethical responsibility.



INNOVATION BY FACULTY RESEARCH AND DEVELOPMENT

FACULTY SEED MONEY PROJECTS

Our Faculty has been consistently engaged in the pursuit of their own independent deliberations by way of doing research. Our Faculty consistently work towards the development of inter-linkages across teaching and research.

Sl. No	Name of the Faculty	Department	Sanctioned Amount (INR in Lakhs)	Year of Sanctioned
1.	Capt. Nazer Hussain	Nautical Science	24000	2025
2	Capt. Manjul Tripathi	Nautical Science	25500	2025

3	Capt. S. Gajendran	Nautical Science	30000	2025
4	Capt. B. Santhakumar	Nautical Science	28500	2025
5	Capt.Karthik Mr.Prathwin Rai	Nautical Science	0.22	2017
6	Capt.Karthik and Varun Rajeevan	Nautical Science	0.3	2017